

Project Title :

Customer behavior Analysis Project Report

using Python, Sql, Powerbi, MS Excel

Introduction

Purchase analysis is the process of examining customer buying behavior, product sales, payment methods, and customer preferences using data analysis techniques.

This project uses SQL queries to analyze customer purchase data stored in a database table named mytable.

The main purpose of this project is to identify purchasing trends, customer interests, and business insights that help improve sales and customer satisfaction.

Objective of the Project

- To analyze customer purchase behavior
- To identify the most purchased clothing items
- To find which age group buys more products
- To analyze gender-wise purchasing patterns
- To identify the most preferred payment method
- To analyze the impact of discounts on customer reviews
- To improve business decision-making using SQL analysis

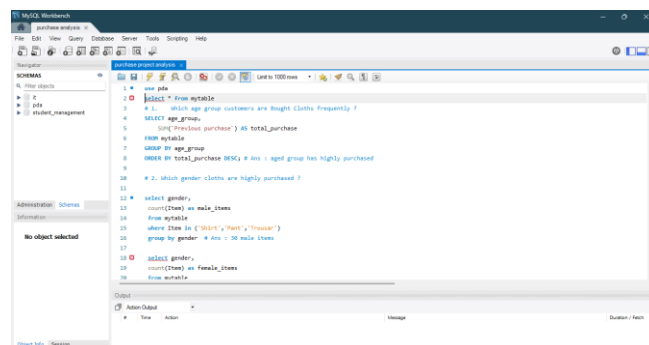
Database Used

Database Name: pda

Table Name: mytable

The table contains customer-related details such as:

- Age
- Gender
- Item Purchased
- Size
- Payment Method
- Discount
- Review



- Previous Purchase

SQL Queries and Analysis

1. Which age group customers bought clothes frequently?

Query

```
SELECT age_group,  
SUM(`Previous purchase`) AS total_purchase  
FROM mytable  
GROUP BY age_group  
ORDER BY total_purchase DESC;
```

Result

The aged group customers purchased clothes more frequently than other age groups.

Explanation

This query groups customers by age category and calculates total purchases made by each group.

	age_group	total_purchase
▶	aged	60
	adult	53
	youth	46

2. Which gender purchased clothes highly?

	gender	male_items
▶	Male	22
	Female	8

Query for Male Items

```
SELECT gender,  
COUNT(Item) AS male_items  
FROM mytable  
WHERE Item IN ('Shirt','Pant','Trousar')  
GROUP BY gender;
```

Query for Female Items

```
SELECT gender,  
COUNT(Item) AS female_items  
FROM mytable  
WHERE Item IN ('Saree', 'Blowse', 'Shorts')  
GROUP BY gender;
```

Result

Male customers purchased more clothing items compared to female customers.

Explanation

The query counts the number of clothing items purchased based on gender.

Advantages of the Project

- Helps understand customer behavior
- Improves sales strategy
- Identifies popular products
- Helps businesses manage inventory

Business Recommendations

Based on the purchase analysis project, the following business recommendations can help improve sales, customer satisfaction, and business growth.

1. Focus on High-Demand Products
2. Target the Most Active Age Group
3. Improve Discount Strategies
4. Optimize Inventory Management
5. Enhance Payment Options
6. Use Data Analytics Regularly

Conclusion

This project successfully analyzed customer purchase data using SQL queries. The analysis identified customer buying patterns, preferred clothing items, payment methods, and the impact of discounts on customer reviews.

Using SQL for purchase analysis helps businesses make better decisions, improve customer experience, and increase sales performance.